

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

10

10 answers · Rationale-rich · Teach from doc

Q1**B**

- B.** multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.

Choice B is the best answer because it describes data from the graph that complete the text's discussion of lizard species' use of maximal speed when escaping predators. According to the text, moving at maximal speed (the highest speed possible) requires so much energy that it is not always an effective strategy for animals, even when they are escaping predators. The graph displays data on the average percent of maximal speed used by lizard species while either escaping predators or pursuing prey. The graph categorizes the data for both pursuing and escaping by the number of species using 30%–39% of maximal speed, 40%–49% of maximal speed, 50%–59% of maximal speed, 60%–69% of maximal speed, 70%–79% of maximal speed, 80%–89% of maximal speed, and 90%–100% of maximal speed, respectively. In the graph, there is at least one species in each of the following percent categories for maximal speed while escaping predators: 50%–59%, 60%–69%, 70%–79%, and 80%–89%. Thus, the data in the graph show that multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.

Choice A is incorrect because the data in the graph isn't organized in such a way that a comparison of the percentage of maximal speed used when escaping predation with the percentage used when pursuing prey is possible at the level of individual species. Choice C is incorrect. It is true that in the graph, the percent category with the largest number of species using maximal speed while escaping predators is 90%–100% (8 species total). However, these data don't complete the text, which is concerned instead with how animals are discouraged from using maximal speed even when escaping predators because of the amount of energy required to use it. Choice D is incorrect because these data from the graph pertain to maximal speed while pursuing prey and therefore don't complete the text's discussion of lizard species' use of maximal speed when escaping predators.

Q2

A

- A. those people who had the right to vote most likely had substantial holdings of French real estate.
-

Choice A is the best answer. We're told that people needed to pay "at least 300 francs in direct taxes" to be able to vote. We're also told that, while "relatively few people paid the tax on real estate," real estate taxes were both the main way people qualified to vote and the main source of revenue for the government. Based on this, we can infer that those who did qualify to vote likely had significant French real estate holdings.

Choice B is incorrect. The passage doesn't mention the voting habits of artisans and merchants nor any reduction in tax burdens on businesses, so there's no basis for this inference. Choice C is incorrect. Although we know that doors and windows were taxed during the Bourbon Restoration, we don't have enough information to infer if doors and windows increased after this time. Choice D is incorrect. Although we know that foreign investments were only minimally taxed, we don't have enough information to determine if those with significant foreign investments were unlikely to have voting rights. For example, it's possible that those with significant foreign investments were likely to also be people with significant domestic investments which they did pay taxes on, so we don't have the information necessary to make this inference.

Q3

C

- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
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Choice C is the best answer because it presents a conclusion about the role of fynbos roots in maintaining the border with the Afro-temperate forest that can be reasonably inferred from the text. The text states that plants in the fynbos have "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." The text then describes an experiment where forest tree seedlings grown in plots isolated from fynbos roots grew at a rate five times greater than seedlings in areas near fynbos roots. This strongly suggests that fynbos roots are depleting soil nutrients to such an extent that forest plants can't get enough nutrients from that soil to flourish, thereby hindering the formation of any significant "transition space" between the habitats and helping keep the border between them sharp.

Choice A is incorrect because there is no evidence in the text that fynbos roots physically damage the root systems of forest plants. Instead, the text explicitly describes fynbos plants as having "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." This description establishes that fynbos roots are specialized for efficient nutrient absorption. The experiment then confirms this function of the roots by showing that forest seedlings "isolated from the roots of fynbos plants exhibited a growth rate five times greater" than those near fynbos roots. This result demonstrates that when forest seedlings don't have to compete with the nutrient-absorbing fynbos roots, they grow much better. Choice B is incorrect. Although the text does indicate that the fynbos and the Afro-temperate forest are distinct from, and adjacent to, each other and that there is "almost no transition space" (area of mixed fynbos and forest plants), the experiment shows that forest plants can grow in soil occupied by fynbos roots, just not very well. So, the idea that the fynbos roots physically obstruct forest plants is unsupported by the text. Choice D is incorrect because it directly contradicts the text's claim that fynbos roots absorb nutrients from, and therefore deplete, the soil they're in, and furthermore, nothing in the text suggests that the habitat of the fynbos is harsh in general.

Q4

D

- D. Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure and allow for outgassing.
-

Choice D is the best answer because it presents findings that, if true, would support the claim made by Bergner and colleagues that the nongravitational acceleration of ‘Oumuamua is due to the expulsion of entrapped hydrogen. The text first introduces the observation of a unique interstellar object named ‘Oumuamua and goes on to explain that the object exhibited nongravitational acceleration that could not be fully attributed to the expected cause: gravitational pull of nearby celestial bodies. The text concludes by stating that Bergner and colleagues claim that the nongravitational acceleration is caused by expulsion of hydrogen gas from ‘Oumuamua’s water-rich icy body. To support this claim requires evidence that hydrogen gas could be present within ‘Oumuamua at all, which this answer choice presents: cosmic radiation can result in embedded pockets of hydrogen gas in water ice. Additionally, evidence that this gas can be released from such a body is required to fully support the claim, which this answer choice goes on to provide: ‘Oumuamua experienced sufficient warming as it traveled through the solar system to alter its icy structure and release the hydrogen gas. Thus, this answer choice provides the best evidence to support Bergner and colleagues’ claim.

Choice A is incorrect because this answer choice concerns faults with previous models of outgassing from ‘Oumuamua of carbon monoxide and nitrogen, which would not support a claim regarding hydrogen outgassing. Furthermore, inconsistencies in other models would not provide evidence in support of a different model or explanation. Choice B is incorrect. The evidence presented in this answer choice would weaken the claim proposed by Bergner and colleagues that the nongravitational acceleration of ‘Oumuamua is caused by the expulsion of hydrogen gas because this answer choice suggests that there is evidence that refutes this claim: ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration caused by the release of hydrogen gas. Furthermore, the remaining portion of this answer choice is unrelated to the claim. Choice C is incorrect because the claim being made by Bergner and colleagues concerns the expulsion of entrapped hydrogen gas, but this answer choice is concerned solely with the differences in magnitude of gravitational and nongravitational acceleration, which would not support Bergner’s claim. Furthermore, this answer choice discusses interstellar objects similar to ‘Oumuamua, but the text states that ‘Oumuamua is the first observed object of its kind in our solar system, so evidence from other, similar bodies would not be available.

Q5

C

- C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.
-

Choice C is the best answer because it most logically completes the text. The text explains that while desert moss species need sufficient sunlight for photosynthesis, exposure to sunlight can also affect the plants negatively by drying them out. Ekwealor and Fisher's team found specimens of *S. caninervis* moss growing under quartz crystals that were semitransparent, allowing some but not all sunlight to pass through them, and the shoot tissue of these specimens was longer than that of *S. caninervis* specimens found growing on the surface that were unprotected by such crystals. Since, as the text explains, the length of shoot tissue is an indicator of plant growth, and since greater exposure to sunlight results in greater loss of moisture, it can be inferred that *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.

Choice A is incorrect because the text doesn't mention another moss species besides *S. caninervis* or discuss whether other such species are able to grow under transparent crystals, as *S. caninervis* is. Choice B is incorrect because, as the text explains, specimens of *S. caninervis* were found growing under quartz crystals and exhibited more plant growth than specimens growing on the soil surface. This wouldn't have been the case if the crystals transmitted insufficient light for the moss's photosynthesis. Choice D is incorrect. The text contrasts the growth of *S. caninervis* specimens found beneath semitransparent quartz crystals with that of specimens found on the soil surface absent such crystals, but it doesn't make a comparison among specimens growing beneath crystals of different degrees of thickness or consider how the relative thickness of the crystals affects the growth of the moss.

Q6

A

- A. The tranquility of a port in the evening can incline a person to appreciate the stillness of old age.
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Choice A is the best answer because it best states the main idea of the text. The speaker describes an evening scene observed from a boat: "little lights in little ports come out" and are reflected in the water's surface "with the stars," fishing boats are moored, "sails are furled" (rolled up), and the riding-light is shining. The speaker emphasizes the calmness of the scene, noting that the "race of wind is stilled" and the atmosphere is sleepy ("slumbrous waters pool a slumbrous world"), and then indicates that the calmness prompts thoughts of how old age might be "sweet" and "quiet" and "gracious in retreat" after "a windy youth" with "storms." Thus, the main idea is that the tranquility of a port in the evening can lead a person to appreciate the potential stillness of old age.

Choice B is incorrect because the speaker isn't discussing the difficulty of bringing a ship into port but describing the calmness of the port at night. And rather than suggesting that old age involves struggles, the speaker suggests that after "a windy youth" with "storms" to be weathered, old age will be "sweet," "quiet," and "gracious"—that is, old age will be tranquil like a port at night. Choice C is incorrect because the speaker isn't suggesting that a person who has lived a varied and active life might find it difficult to stay in a calm place in old age. On the contrary, the speaker suggests that after a "windy youth" with "storms," old age can be welcomed as something "sweet," "quiet," and "gracious," just like a calm port at night. Choice D is incorrect. Although the speaker does touch on the idea of a "windy youth" followed by "sweet old age," the speaker doesn't tie that idea to a contrast between daytime activity and nighttime peacefulness at a port. The speaker doesn't address any activity during the day at a port, only briefly mentioning that the "race of wind is stilled" at the end of the day, and instead focuses on the tranquility of the port at night and how it prompts thoughts of "quiet" and "gracious" old age.

Q7

A

- A. The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.
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Choice A is the best answer because it most effectively uses data from the table to support the student's argument about the role of bufadienolide in the egg preferences of cane toad tadpoles. For each of five amphibian species included in the 2022 study, the table gives the percentage of available eggs that the cane toad tadpoles ate. According to the table, the tadpoles ate 10% of striped burrowing frog eggs and 1% of dainty green tree frog eggs, which suggests a preference for striped burrowing frog eggs over dainty green tree frog eggs. The table also indicates that neither of these species' eggs produces bufadienolide. Thus, these data suggest that something other than the presence or absence of bufadienolide is needed to adequately explain the tadpoles' egg preferences.

Choice B is incorrect. Although the table shows that for each of the five amphibian species, the cane toad tadpoles ate less than 100% of that species' eggs, which demonstrates that the tadpoles did indeed leave some eggs for each species unharmed, this fact alone is irrelevant to the tadpoles' preferences for some species' eggs over other species' eggs. Choice C is incorrect. Although the table indicates that the cane toad tadpoles ate 90% of the cane toad eggs and 7% of the short-footed frog eggs, which suggests that they prefer cane toad eggs over short-footed frog eggs, the table also indicates that cane toad eggs produce bufadienolide, whereas short-footed frog eggs do not. Therefore, these data are not sufficient to exclude that bufadienolide alone could explain the tadpoles' preference for some species' eggs over other species' eggs. Choice D is incorrect. Although the table shows that for both dainty green tree frog eggs and little red tree frog eggs, the cane toad tadpoles ate 1% of those species' eggs, it also indicates that neither produces bufadienolide. Thus, these data alone don't indicate bufadienolide's role in the tadpoles' egg preferences.

Q8

B

- B. need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.
-

Choice B is the best answer. The text describes a study examining the relationship between a species of parasitic tapeworm, *A. brevis*, and its host insect, the *T. nylander* ant. According to the text, researchers were surprised to find that the tapeworm extends the life of its ant host, rather than reducing it. The text goes on to state that the infected ants end up doing less work to sustain the colony and that as a result, the uninfected ants take on the infected ants' share of labor in addition to caring for them in their infected state. The study's researchers also observed that the uninfected ants have shorter lifespans than expected. If the infected ants, who are doing less work in the colony, have longer lifespans, it can be inferred that the less an ant works, the longer it will live. The opposite of this statement can also be inferred: the more an ant works, the shorter its life. So, since the workload within the colony is being redistributed so that the infected ants work less while the uninfected ants work more (as they take on the neglected duties of the infected ants and also care for those ants), then it can be inferred that the lifespans of the uninfected ants are shortened because the need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

Choice A is incorrect because the text does not indicate how *A. brevis* is transmitted to the ants or assert that uninfected ants are more likely to be directly exposed to *A. brevis* while caring for infected ants. Choice C is incorrect because the text makes no mention of the relative abilities of infected and uninfected ants to escape predators: in fact, predators are not mentioned in the text at all. Choice D is incorrect because the text does not supply any information about the average lifespans of the ants in colonies without parasitic activity; the text only indicates factors that lengthen and shorten the lifespans of ants in parasitized colonies.

Q9

C

- C. The production’s dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.
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Choice C is the best answer. The text says that the production’s use of its large, winding space has a very specific effect on the audience. Given that the space itself is so important to creating this effect, it would be difficult to reproduce the production in a different space.

Choice A is incorrect. The fact that the venue is in New York City isn’t connected to the experimental nature of the performance. It’s the size of the venue, not its location in New York, that affects the theatrical experience. Choice B is incorrect. The text never suggests that audience members are disappointed because they can’t see the entire production. In fact, it suggests that that’s an important part of the experience. Choice D is incorrect. The text doesn’t mention a recommended itinerary for audience members.

Q10

C

- C. “Lumumba’s practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention.”
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Choice C is the best answer because it illustrates the student’s claim about some historians viewing Lumumba primarily as a symbol. This quotation argues that Lumumba “warrants” (or deserves) “scholarly attention” as a symbol and not for his “practical accomplishments”—that is, his actions as prime minister—which “can be passed over quickly,” or dismissed as being of comparatively little importance. Thus, the quotation expresses the view that the student criticizes some historians for holding.

Choice A is incorrect. Although this quotation touches on the difficulty of evaluating Lumumba’s legacy, it doesn’t address how historians of modern African politics view him as a symbol. Choice B is incorrect. While this quotation mentions Lumumba’s political beliefs, it doesn’t discuss historians viewing him as a symbol. Choice D is incorrect. This quotation touches on Lumumba’s vision for his country, but it doesn’t discuss historians viewing him as a symbol.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

